

A three-stage intelligent decision framework for post-disaster search and rescue open-loop routing

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ABSTRACT

Search and Rescue (SAR) path planning is critical in post-disaster scenarios with road damage and noisy demand information. This paper proposes a dynamic post-disaster SAR path planning model that comprehensively considers road damage severity, casualty uncertainty, and life-saving priority. A three-stage decision-making framework is developed: first, an improved K-means clustering algorithm based on silhouette analysis determines the optimal number of rescue clusters and establishes an unstructured post-disaster environment model. Second, SAR path pre-planning is implemented: a hybrid heuristic integrating priority greedy hierarchical allocation, priority-aware nearest neighbor heuristic and constraint-preserving 2-opt algorithm generates high-quality initial solutions, and a modified dynamic adaptive large neighborhood search (M-D-ALNS) algorithm constructs pre-planned SAR routes with minimal priority violations and shortest completion time. Third, a bidirectional event-driven response mechanism based on local and global (BEDRM-LG) is proposed to mitigate discrepancies between pre-planned results and actual rescue conditions. Experimental validation on modified Solomon benchmark instances shows the proposed initialization strategy generates optimal initial solutions fastest; M-D-ALNS reduces violation rate to near zero with over 90% optimization efficiency; BEDRM-LG outperforms common benchmarks and even reaches the theoretical upper bound, demonstrating superior performance over static planning and continuous full-information re-optimization in post-disaster SAR tasks.

1. Introduction

Large-scale natural disasters, such as earthquakes and typhoons, often cause catastrophic damage to infrastructure and result in mass casualties within minutes. In the immediate aftermath, the efficiency of Search and Rescue (SAR) operations during the "Golden 72 Hours" is the decisive factor in reducing mortality rates [1,2]. Prior to the deployment of rescue teams, advanced reconnaissance and assessment are essential. Recent studies have leveraged deep learning, such as Vision Transformers (ViT) [3], to achieve fine-grained earthquake structural damage detection, providing a critical foundation for disaster verification. However, translating these assessments into effective rescue actions remains a challenge. SAR teams must still navigate a highly dynamic environment: roads may be blocked by debris, and the number of victims is initially unknown or estimated with high noise [4].

Post-disaster SAR routing differs fundamentally from commercial Vehicle Routing Problems (VRP) in two critical aspects. First, the

optimization objectives need to satisfy humanitarian and rescue efficiency requirements, namely minimizing the makespan (time to rescue the last victim) and prioritizing critical zones, rather than minimizing economic costs [5,6]. Second, rescue operations often require an open-loop structure. Unlike delivery trucks that return to a central depot, rescue vehicles in heavily damaged areas often proceed directly to hospitals, safe havens, or subsequent disaster sites to maximize time utilization [7].

Emerging research has begun to address these complexities using IoT-enabled autonomous systems. For instance, some frameworks optimize UAV path planning for green IoT systems to ensure energy-efficient data collection via energy harvesting [8], while others integrate Deep Reinforcement Learning (DRL) with kinematic optimization to enable real-time obstacle avoidance and survivor search in collapsing urban environments [9]. Although DRL-based models achieve excellent computational speed for real-time local navigation and reconnaissance, they often focus on the pre-rescue "information preparation" stage. By

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contrast, grounded SAR operations involve high-dimensional logistics problems with strict humanitarian constraints, where traditional DRL typically faces training instability and complex constraint-handling difficulties.

Despite these advancements, existing SAR research largely relies on static planning or assumes perfect information availability [10,11]. Furthermore, while IoT-enabled systems emphasize real-time data fusion, the risk of information overload and overfitting to high-frequency noise in chaotic environments is often overlooked. While some Dynamic Vehicle Routing Problems (DVRP) approaches address uncertainty, they often lack specific mechanisms to handle the trade-off between exploring uncertain demands and exploiting known high-priority zones [12]. Traditional "rolling horizon" methods may trigger excessive re-planning computations or, conversely, fail to react swiftly to "Black Swan" events (e.g., sudden bridge collapses) [13,14].

To address these gaps, this paper proposes a Three-Stage Intelligent Decision Framework for the Post-Disaster Dynamic Search and Rescue Open-loop Routing Problem. Unlike the model that focuses on searching for the trapped individuals, our study focuses on the execution of multi-team rescue logistics under deep uncertainty. The framework we have proposed integrates priority optimization with search-and-rescue task allocation and scheduling to address humanitarian relief demands and efficiency requirements in uncertain environments. The main contributions of this study are summarized as follows:

- **Realistic Post-Disaster Environment Modeling:** We construct rigorous spatial correlation road damage models and demand disturbance models to simulate unstructured post-disaster environments, bridging the gap between theoretical vehicle routing problems and actual disaster scenarios.
- **A Hierarchical Decision Framework:** We propose a three-stage decision-making framework for post-earthquake uncertain environments, namely hierarchical allocation-pre-planning-dynamic adjustment, to decompose complex dynamic problems. An improved K-means clustering method is adopted to achieve adaptive optimal region partitioning, and a multi-indicator priority evaluation system is constructed with weights objectively determined using the Entropy Weight Method (EWM).
- **Priority-Aware Algorithm:** We develop a hybrid heuristic initial solution generation strategy for multi-rescue-station scenarios, where a priority-aware nearest neighbor algorithm and constraint-preserving 2-opt local search are adopted to generate highly feasible initial solutions. Furthermore, an improved adaptive large neighborhood search algorithm incorporated with specialized operators is developed to further reduce constraint violations and shorten rescue makespan.
- **Event-Driven Dynamic Mechanism:** We designed an impact-based agile triggering mechanism and demonstrated that, under time constraints for emergency response in highly uncertain post-disaster environments, the robustness of agile responses may outperform perfect planning from an omniscient perspective.

2. Related works

2.1. Metaheuristic for post-disaster open routing

Post-disaster SAR routing is technically modeled as a variant of the Open Vehicle Routing Problem (OVRP). Unlike commercial logistics where vehicles form closed loops returning to the depot, rescue vehicles in the "Golden 72 Hours" must maximize coverage by ending their routes at the last rescue site or a nearest safe haven. This open-loop characteristic significantly expands the search space and prevents the direct application of standard TSP/VRP heuristics.

Given that VRP and its variants are NP-hard, exact methods are computationally prohibitive for large-scale disaster instances. Consequently, metaheuristic have become the dominant solution approach

[15,16]. Recent comprehensive surveys indicate a growing trend towards hybrid metaheuristic that combine population-based global search with trajectory-based local refinement to handle complex constraints [17,18].

Among these, Adaptive Large Neighborhood Search (ALNS) has proven exceptionally robust. By dynamically selecting from a portfolio of destruction and repair operators, ALNS can effectively escape local optima, a critical feature when the solution landscape is rugged due to road blockages [19]. Recent studies have successfully adapted ALNS for rich VRP variants, such as sustainable routing [20] and heterogeneous fleet management [21,22]. Notably, Goodarzian et al. [23] demonstrated the efficacy of hybrid heuristics for medical supply chains during crises. However, a significant gap remains: most existing ALNS implementations treat "routing" and "prioritization" as separate sequential phases. Traditional operators rely heavily on distance-based insertion [24], which often fails to capture the urgency of life-saving priorities. This study innovates by embedding "priority-aware" mechanisms directly into the ALNS operator library.

2.2. Uncertainty modeling and dynamic adaptation

Disaster relief operations are inherently plagued by deep uncertainty, particularly regarding road network connectivity and the precise number of casualties [25]. Similar to Liu et al. [26], in the path planning process, we adopted a two-stage logic, but further incorporated road damage and personnel uncertainty to better reflect the actual disaster situation. Jeong et al. used Monte Carlo simulation to address the "spatial drift uncertainty" caused by ocean current wind directions, which also inspired us to use Monte Carlo to validate scenarios of road spatial damage and casualties [27].

Some research addressed these uncertainties through static robust optimization, such as two-stage stochastic programming [28] or robust planning models [10]. While these methods provide mathematically guaranteed bounds, they are often too conservative and computationally expensive for real-time application. The focus has thus shifted to Dynamic VRP strategies that leverage real-time information updates [29]. Ouertani et al. systematically reviewed DVRP and highlighted that the core of dynamic routing lies in re-optimization with newly emerging information within limited time [30]; Abbatecola et al. comprehensively summarized advanced DVRP solution methods [31]. Subsequent studies classified dynamic events and analyzed the applicability of periodic re-planning [32] and rolling horizon re-planning [29]. Some works also adopted multi-stage models or rolling time windows for multi-period optimization of demand arrivals and resource updates [33]. Our framework adopts an impact-based trigger mechanism, balancing information accuracy, rescue priorities, and rescue timeliness. We distinguish between "estimated victims" and "actual victims", setting a dynamic trigger threshold that prioritizes life-saving needs and maximizes re-planning benefits over mere travel cost reductions.

2.3. Data-driven prioritization via MCDM

Humanitarian operations require balancing conflicting objectives, such as efficiency (makespan), equity, and urgency (priority) [5]. In the chaotic initial phase of a disaster, determining which affected areas to visit first is a complex Multi-Criteria Decision Making (MCDM) problem.

Traditional methods like AHP and TOPSIS rely heavily on expert judgment matrices [34]. However, the immediate post-disaster stage is usually time-critical, expert opinions may diverge considerably, and subjective judgments are prone to cognitive bias under stress. To mitigate the influence of subjective factors, objective weighting methods such as the Entropy Weight Method and the Standard Deviation Method (EWM) have gained increasing attention in recent years [35]. EWM has gained traction for its ability to quantify the information entropy of indicators, effectively assigning higher weights to factors with greater variability and discriminative power [36]. Zhang et al. [21] recently

applied dynamic entropy weighting to maritime emergency decision-making, validating its effectiveness in time-critical environments.

Drawing on the studies of Xia et al. [37] and Huang et al. [38], we integrate EWM with spatial clustering and derive objective priority scores based on indicators including Total Life Value (TLV), Disaster Impact (DI), and Rescue Efficiency Density. This links assessment with routing, ensuring that the generated routes are not only geographically efficient but also ethically aligned with the imperative of saving lives.

3. Problem formulation

To describe the post-disaster rescue task clearly, we define the Post-Disaster Dynamic Rescue Routing Problem (PD-DRRP) as a special open vehicle routing problem with priorities and dynamic information, as shown in Fig. 1. The PD-DRRP has three core features: (1) The road network is damaged, and travel times are heterogeneous and stochastic; (2) Rescue demand at each site is uncertain, with only noisy estimates available at the planning stage; (3) There is an explicit priority order between regions, which routes should follow as much as possible, and the system can perform re-planning once ground truth information is revealed.

3.1. System definition and notation

Let $G = (V, E)$ be a complete graph representing the rescue area, where $V = D \cup N$ is the set of nodes. Here, $D = \{d_1, d_2, \dots, d_t\} (t \geq 1)$ represents the set of multiple rescue centers. Node 0 represents the epicenter and is only used to construct damage zones. $N = \{1, 2, \dots, n\}$ is the set of discrete demand points requiring rescue. A fleet of m homogeneous rescue teams, denoted by $K = \{1, 2, \dots, m\}$, is available. Each team $k \in K$ departs from an initial rescue station $s_k \in D$ and completes its route upon returning to any nearest rescue station $e_k \in D$ after finishing all search and rescue missions. The definitions of model symbols are presented in Table 1.

3.2. Objective and constraints

In disaster rescue, the main goal is to finish all tasks as early as possible (makespan) rather than minimizing distance. The primary

Table 1
PD-DRRP model variable definitions.

Symbol	Meaning
V	Set of all nodes, $V = D \cup N$
D	Set of multiple depots, $D = \{d_1, d_2, \dots, d_t\} (t \geq 1)$
N	Set of demand points, $N = \{1, 2, \dots, n\}$
K	Set of rescue teams, $K = \{1, 2, \dots, m\}$
C_{ij}	Travel time from node i to node j
s_i^{est}	The estimated search and rescue service time for point i
s_i^{act}	The actual search and rescue service time for point i
M	A sufficiently large positive number.
τ	Current re-planning epoch. $\{\tau_0, \tau_1, \dots\}$ are a series of decision epochs.
$N_{pen}(\tau)$	At time τ , the set of rescue points that have not yet begun to be served and require reallocation and re-planning is denoted as $N_{pending}$.
x_{ijk}	1 if rescue team k travels directly from node i to node j , 0 otherwise
u_{ik}	Sequence number of node i in the route of rescue team k
z_{ks}	1 if rescue team k departs from depot s , 0 otherwise
w_{ke}	1 if rescue team k returns to depot e to end its route, 0 otherwise
δ_{ij}	1 if node i has a lower access priority than node j , 0 otherwise.
y_{ik}^{τ}	1 if task i is assigned to team k in the new plan, 0 otherwise.
y_{ik}^{old}	1 if task i was assigned to team k in the previous plan, 0 otherwise.

objective is defined as:

$$\Pi_1 = \min \left(\max_{k \in K} T_k \right) \quad (1)$$

where T_k is the total operational time for team k , composed of travel time and service time:

$$T_k = \sum_{i \in V} \sum_{j \in V} C_{ij} \cdot x_{ijk} + \sum_{i \in N_k} s_i^{est} \quad (2)$$

For any task node i , s_i^{est} is determined by the product of the estimated number of trapped people and the average rescue time per person.

Let PS_i denote the objective priority score of node i (where a larger score indicates higher urgency, as calculated in Section 4.1.2). The number of priority violations for team k is formulated as:

$$V_k = \sum_{i \in V} \sum_{j \in V} x_{ijk} \cdot \delta_{ij} \quad (3)$$

Whenever a rescue team travels directly from a lower-priority node to a higher-priority node, a priority violation is recorded. The total count

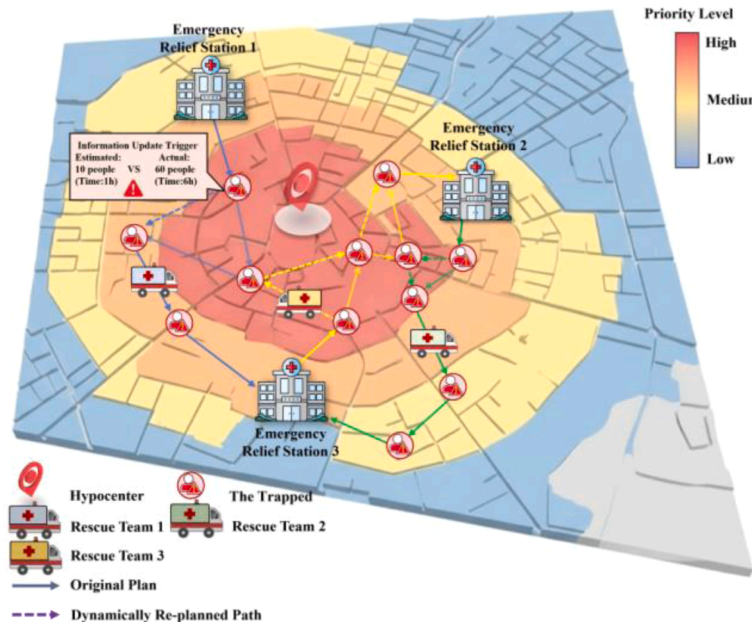


Fig. 1. Illustration of the post-disaster rescue scenario and dynamic re-planning.

of such violations yields the team's total priority violations V_k . If the priority scores of consecutively visited nodes strictly decrease along a path, zero violations are achieved, indicating that the rescue plan fully complies with humanitarian emergency principles. To handle soft constraints (priority violations), we introduce a penalty term into the objective function evaluated during optimization:

$$\Pi_2 = \min \left(\max_{k \in K} T_k + M \cdot \sum_{k \in K} V_k \right) \quad (4)$$

Among them, the penalty coefficient M is set to 1000 after experimental validation, with the experimental results shown in Table S1.

The PD-DRRP model adheres to the following constraints:

$$\sum_{s \in D} z_{ks} = 1, \quad \forall k \in K \quad (5)$$

$$\sum_{e \in D} w_{ek} = 1, \quad \forall k \in K \quad (6)$$

$$\sum_{j \in V \setminus \{s\}} x_{sjk} = z_{sk}, \quad \forall s \in D, \forall k \in K \quad (7)$$

$$\sum_{i \in V \setminus \{e\}} x_{iek} = w_{ek}, \quad \forall e \in D, \forall k \in K \quad (8)$$

$$\sum_{k \in K} \sum_{j \in V \setminus \{i\}} x_{ijk} = 1, \quad \forall i \in N \quad (9)$$

$$\sum_{j \in V \setminus \{i\}} x_{ijk} = \sum_{j \in V \setminus \{i\}} x_{jik}, \quad \forall i \in N, \forall k \in K \quad (10)$$

$$u_{sk} = 1, \quad \forall k \in K \quad (11)$$

$$u_{jk} \geq u_{ik} + 1 - M(1 - x_{ijk}), \quad \forall i, j \in V, i \neq j, \forall k \in K \quad (12)$$

$$x_{ijk}, z_{sk}, w_{ek}, \delta_{ij}, y_{ik}^r, y_{ik}^{old} \in \{0, 1\} \quad \forall i, j \in V, \forall k \in K \quad (13)$$

$$1 \leq u_{ik} \leq n \quad \forall i \in V, \forall k \in K \quad (14)$$

Eqs. (5) and (6) ensure each team departs from one depot and ends at one depot. Eqs. (7) and (8) link the depot selection variables to the routing variables. Eq. (9) ensures every demand node is visited exactly once. Eq. (10) guarantees flow conservation. Eq. (12) is the Miller-Tucker-Zemlin (MTZ) formulation for sub-tour elimination. Eqs. (13) and (14) define the domain of the decision variables.

During the dynamic re-planning stage, the objective function is revised to minimize the maximum search completion time after the decision-making moment τ and the penalty for constraint violations. To avoid confusion and fatigue among on-site rescuers caused by frequent and drastic plan changes, an adjustment ratio threshold ψ is introduced. The deviation between the new plan $\mathcal{P}_{new}$ and the previous plan $\mathcal{P}_{old}$ is defined using the Hamming distance between assignment variables. Considering double-counting issues where a single task leaves one team and joins another, the total number of changes in assignment variables is discounted by a coefficient of 2.

$$\sum_{i \in N_{pen}(\tau)} \sum_{k \in K} |y_{ik}^r - y_{ik}^{old}| \leq 2\psi |N_{pen}(\tau)| \quad (15)$$

To mitigate excessive frequent responses within a short period, a rolling horizon mechanism is adopted to regulate re-planning triggers:

Let E denote the set of events that trigger re-planning, with their occurrence time denoted by t_{curr} . A new decision epoch τ_{new} is activated if and only if $t_{curr} \geq \tau_{last} + \Delta t_{min}$. Otherwise, the event is added to the event pool without triggering an immediate re-planning process.

Table S4 summarizes the experimental results for determining the optimal values of the re-planning interval Δt_{min} and stability ratio ψ .

3.3. Post-disaster environment modeling

To rigorously evaluate the proposed framework, we construct a stochastic post-disaster environment that simulates the spatial correlation of infrastructure damage and the uncertainty of demand information.

3.3.1. Spatially correlated road damage

The post-disaster road damage model described in this paper is designed to simulate high-intensity earthquake scenarios, such as those with a magnitude of 7.0 or higher and an epicentral intensity of IX or X, similar to the catastrophic 2008 Wenchuan earthquake. Under such extreme conditions, widespread structural damage, complete road blockages in severely affected areas, and victims being buried under debris are common occurrences. Drawing from seismic damage distribution patterns [39,40], infrastructure damage is highly correlated with the Euclidean distance from the epicenter. We partition the road network into three concentric zones based on the distance d_i from the epicenter to node i .

Let R_{max} denote the maximum coverage radius of the rescue region. The zoning thresholds R_1 and R_2 are defined as proportional fractions of R_{max} :

$$R_1 = 0.2 \times R_{max}, \quad R_2 = 0.5 \times R_{max} \quad (16)$$

The zones are defined as follows:

- Core Zone ($d \leq R_1$): The area adjacent to the epicenter, suffering the most severe structural destruction.
- Severe Zone ($R_1 < d \leq R_2$): The transitional area exhibiting moderate infrastructure damage.
- Marginal Zone ($d > R_2$): The peripheral area with minor impact on mobility.

Drawing upon weighting matrix methodologies utilized in historical seismic casualty and damage assessments [41], the post-disaster travel time C'_{ij} is modeled by applying an impedance coefficient μ to the original travel time C_{ij} . Based on the study by Chen et al. [42], we define three road status levels: (1)Blocked ($\mu = 4.0$), where the road is severed requiring significant detours; (2)Damaged ($\mu = 2.0$), where capacity is reduced due to debris; and (3)Clear ($\mu = 1.1$), representing near-normal conditions. Notably, as highlighted in recent urban seismic resilience evaluations [43], an idealistic multiplier of 1.0 is practically non-existent during the "Golden 72 Hours." Even structurally "clear" roads suffer from surface cracks, scattered rubble, and spontaneous civilian evacuation traffic, making 1.1 a much more realistic baseline for near-normal conditions.

The probability of a road segment falling into a specific category depends on the zone it traverses. Empirical studies demonstrate a strong correlation between epicentral distance, earthquake magnitude, and severe road damage (RDS) [44]. In the immediate Core Zone of a high-magnitude event, the probability of structural road failure approaches absolute certainty. As presented in Table 2, roads in the Core Zone have a high probability of being blocked. The impact of this damage model is visualized in Figs. 2 and 3.

3.3.2. Uncertain rescue demand generation

In the initial phase of a disaster, exact casualty numbers are unavailable. The rescue command center operates based on rough,

Table 2
Probability distribution of road impedance (μ) across zones.

Zone	Blocked ($\times 4.0$)	Damaged ($\times 2.0$)	Clear ($\times 1.1$)
Core Zone	90%	10%	0%
Severe Zone	20%	70%	10%
Marginal Zone	10%	0%	90%

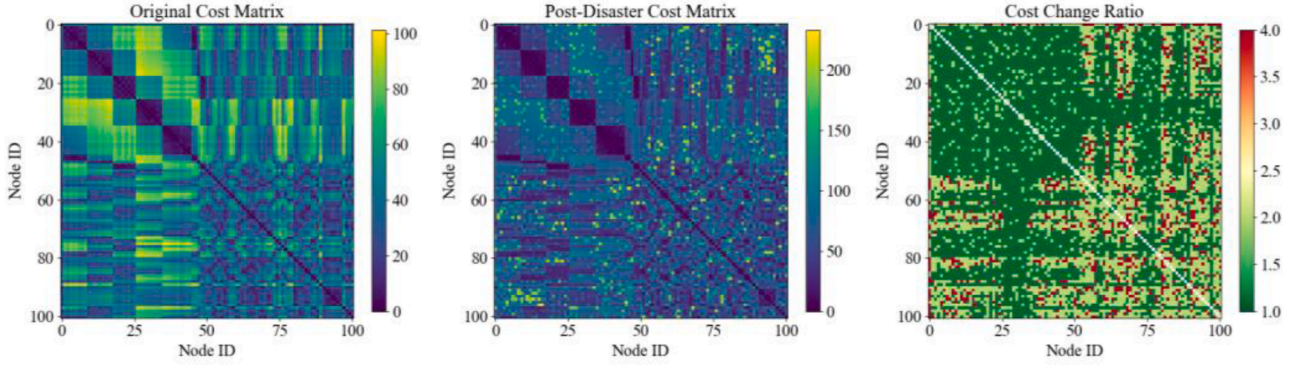


Fig. 2. Road cost matrix three-dimensional view.

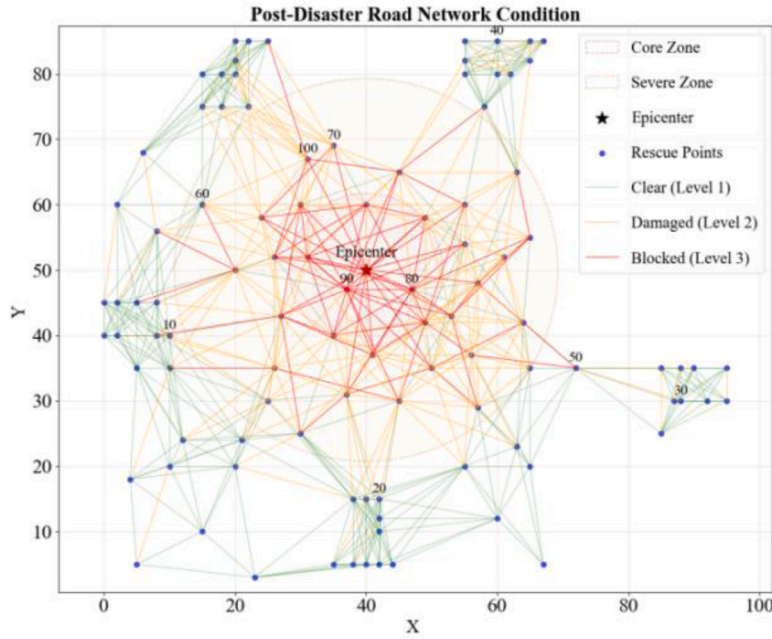


Fig. 3. Visualization of the simulated post-disaster road network.

delayed, and sometimes biased field reports [45]. To simulate this stochasticity scientifically, we adopt a probability distribution-based generation method consistent with robust optimization literature. We distinguish between two datasets:

(1) Actual Victims (L_{act}): We denote the real number of trapped victims at node i by $L_{act}(i)$. We generate this ground truth from a Poisson distribution [46]:

$$L_{act}(i) \sim \text{Poisson}(P_{base}(i)) \quad (17)$$

According to the population exposure model proposed by Ding et al. [47] in their research on the spatio-temporal assessment framework of casualties, the baseline demand $P_{base}(i)$ is not uniform across nodes; it is generated proportionally to the node density of the parent cluster. Nodes located in dense clusters are assigned higher $P_{base}(i)$ values, while isolated nodes receive lower values. This density-based construction is consistent with empirical evidence that areas with higher population density tend to have higher casualty counts and greater demand for relief.

(2) Estimated Victims (L_{est}): Decision makers at $t=0$ only observe an imperfect estimate. We denote it by $L_{est}(i)$ and generate it by adding zero-mean Gaussian noise to the actual victims [47,48]:

$$L_{est}(i) = \max\{0, L_{act}(i) + \varepsilon_i\}, \varepsilon_i \sim N(0, \sigma^2) \quad (18)$$

The variance σ^2 controls the noise level; a larger σ^2 implies more severe information errors. Negative values are truncated at zero to ensure non-negative estimated casualty counts.

Taking the 100 nodes in the constructed instance RC206 as an example, the mean absolute error (MAE) between the estimated and actual population is 3.80, with no extreme deviations overall while preserving the spatial distribution characteristics of risk. The discrepancy is illustrated in Fig. 4.

4. The proposed three-stage intelligent decision framework

Fig. 5 depicts the three-stage decision-making framework developed in this study. Based on post-disaster road network conditions, demand uncertainty, and regional priorities, the framework generates highly feasible emergency rescue plans (with few priority violations and short

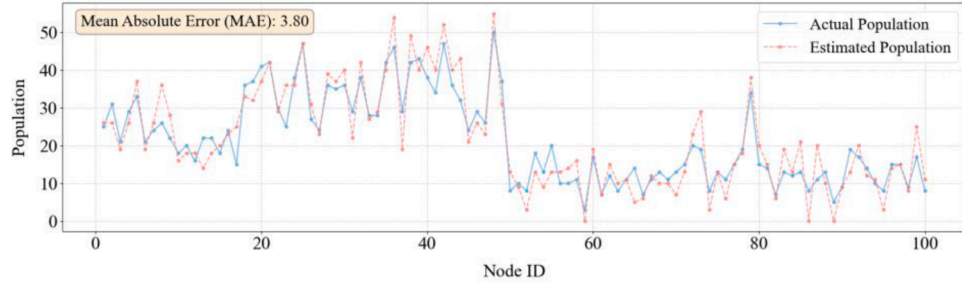


Fig. 4. Comparison between estimated victims (L_{est}) and actual victims (L_{act}) demand.

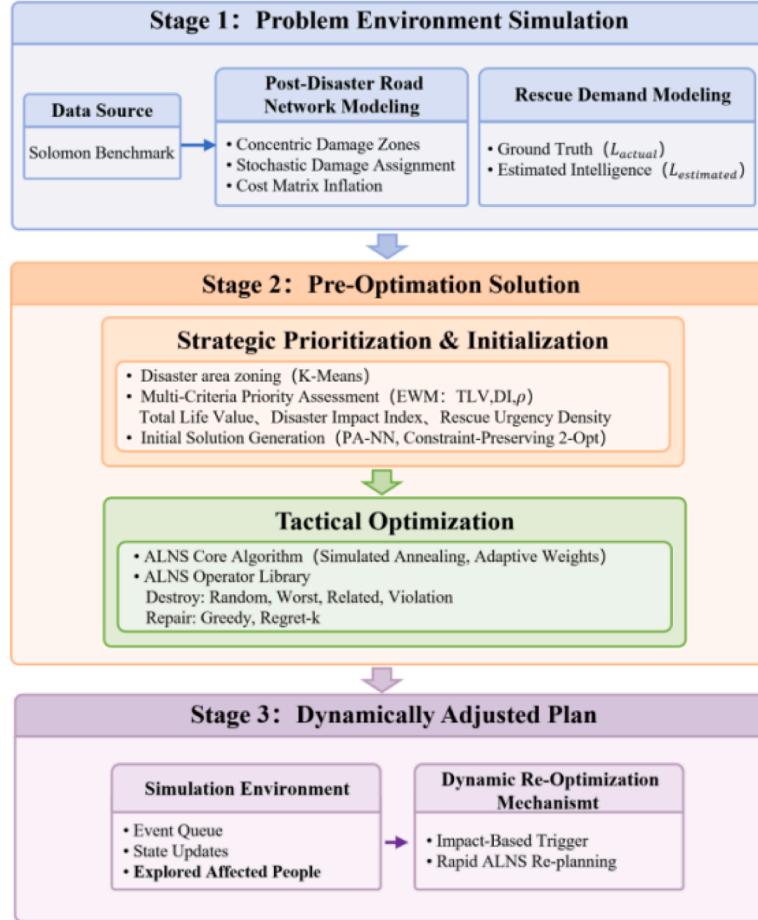


Fig. 5. The three-stage intelligent decision-making framework proposed in this study.

rescue makespans) while enabling rapid responses to dynamic information updates.

4.1. Stage 1: rescue prioritization determination and initialization

To transform a large-scale chaotic disaster area into manageable operational units, we employ a "hierarchical cluster greedy allocation" strategy. This phase decomposes the entire rescue area into multiple task packages and generates a preliminary solution that adheres as strictly as possible to rescue priorities.

4.1.1. Data-driven zoning via optimized clustering

First, the demand nodes N are partitioned into K geographical clusters. The choice of K is critical, an arbitrary choice would introduce subjectivity. To avoid this, we employ Silhouette Analysis to determine

the optimal K in a data-driven way [49]. For a specific node i , the silhouette coefficient $s(i)$ measures the trade-off between cluster tightness and separation:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad (19)$$

- **$a(i)$ (Cohesion):** The mean distance from node i to all other points in the same cluster. A small $a(i)$ implies the node is well-situated within a dense cluster.
- **$b(i)$ (Separation):** The mean distance from node i to all points in the nearest neighboring cluster. A large $b(i)$ signifies clear separation from other zones.

We iterate K from 1 to 30 and calculate the average silhouette co-

efficient for each configuration. As shown in Fig. 6a, for all transformed datasets, we select the K value that yields the local maximum score, balancing cohesion and separability.

4.1.2. Objective priority assessment

Once clustered, we must evaluate the rescue urgency of each zone. Following established practices in post-disaster resource allocation and spatial risk assessment [50,51], we construct three indicators for each cluster based on estimated victims to capture the complexity of the post-disaster environment:

- **Total Life Value (TLV):** The sum of estimated survivors ($TLV_k = \sum_{i \in C_k} L_{est}(i)$), representing the potential humanitarian gain.
- **Disaster Impact Index (DI):** Quantified inversely by the distance to the epicenter. Although distance from the epicenter affects the extent of road damage and travel time, this paper treats DI as an independent indicator to measure medical urgency and structural vulnerability. Areas closer to the epicenter experience higher seismic intensity, making buildings more prone to severe collapse. Victims in these areas tend to suffer more severe injuries and have shorter survival times, which plays a crucial balancing role in the algorithm. Without incorporating DI, a distance-based greedy path algorithm would intentionally avoid the epicenter area because of the severe road damage and higher travel costs there.
- **Rescue Efficiency Density (ρ):** The cluster's TLV divided by its convex hull area, identifying high-density rescue hotspots and reflects the potential efficiency of rescue operations in those areas.

To determine the weights of these indicators objectively, we employ the **Entropy Weight Method (EWM)**. This method assigns weights based on data volatility, avoiding the bias of subjective expert scoring. The calculation steps are:

1. **Standardization:** Normalize the raw indicator matrix X to $Y_{ij} \in [0, 1]$ to eliminate unit difference.
2. **Entropy Calculation:** Compute the information entropy E_j for each indicator j :

$$E_j = k \sum_i p_{ij} \ln p_{ij}, \text{ where } p_{ij} = p_{ij} = \frac{Y_{ij}}{\sum_i Y_{ij}} \quad (20)$$

Here, $k = 1 / \ln(m)$ is the normalization factor for m clusters. Lower entropy implies the indicator carries more useful information.

3. **Weight Determination:** The objective weight w_j is derived from the redundancy $d_j = 1 - E_j$:

$$w_j = \frac{1 - E_j}{\sum_k (1 - E_k)} \quad (21)$$

The final Priority Score for cluster i is calculated as $PS_i = \sum_j w_j Y_{ij}$. Fig. 6b shows the priority calculation results and spatial distribution of clusters for the modified case study RC206.

4.1.3. Greedy hierarchical task allocation and initial routing

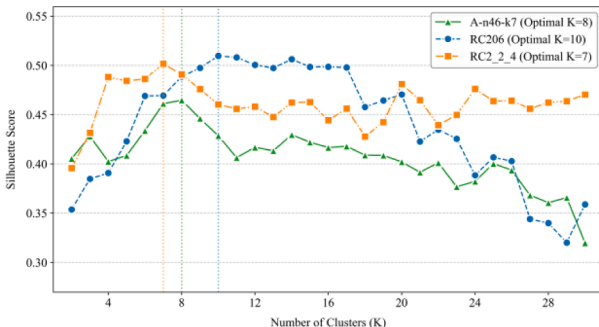
Based on the priority ranking derived above, we generate a highly priority-compliant initial solution through the following three-step procedure.

1. Greedy Hierarchical Allocation (GHA): To balance work-load and priority, we employ a hierarchical allocation strategy. The top m highest-priority clusters are assigned one-to-one to the m rescue teams to serve as their core operational zones. The centroid of each assigned cluster defines the team's starting region. Subsequently, the remaining lower-priority clusters are assigned to the nearest team based on Euclidean distance. This ensures that each team's responsibility area is compact and centered around a high-priority core.

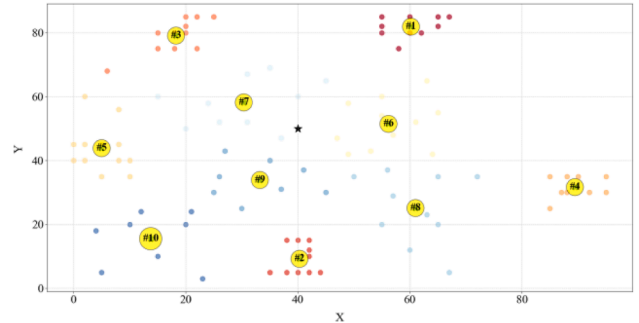
2. Priority-Aware Open Route Construction: Within each assigned zone, we construct the route using a Priority-Aware Nearest Neighbor (PA-NN) heuristic. Unlike the standard nearest algorithm, we dynamically modify the traversal cost to penalize "Priority Inversions". The modified cost $\tilde{C}_{ij}$ is defined as:

$$\tilde{C}_{ij} = \begin{cases} C_{ij} \cdot \omega, & \text{if } PS_i < PS_j \text{ (Low} \rightarrow \text{High)} \\ C_{ij}, & \text{otherwise} \end{cases} \quad (22)$$

Here, a secondary multiplicative weighting mechanism is introduced. the virtual cost $\tilde{C}_{ij}$ is generated by applying a pure algorithmic penalty weight ω (set to 2 in this article) to the path-loss weighted travel cost C_{ij} , which already encodes post-disaster travel impedance via the path loss coefficient μ . This virtual cost guides the nearest-neighbor search of the algorithm. Concretely, this parameter setting means that the algorithm will only violate the humanitarian priority rule when the actual post-disaster travel time to the priority-compliant node exceeds twice that to the non-compliant node. Such cases are extremely rare in compact task areas with well-defined local boundaries. This secondary weighting mechanism effectively directs the algorithm to adhere to humanitarian priorities, while eliminating the substantial increase in total rescue time caused by forced absolute priority sorting—an approach that would otherwise induce massive vehicle detours. Subse-



(a) Determining the optimal clustering k-value



(b) Spatial distribution of the RC206 modification case study

Fig. 6. Optimal clustering k-value for regions and its spatial distribution.

quent experimental validations verify the algorithm's robustness under this configuration: as indicated by the V_{init} metric of the proposed strategy in Table S8, our method yields the lowest initial violation count across all test cases.

3. Constraint-Preserving 2-Opt Refinement: We then apply a 2-opt local search to further shorten the route. This method traversing all pairs of nodes that can be exchanged (i, j) and reversing the $i - j$ section of the path to generate a new path. Crucially, a move is accepted only if the total distance decreases and the number of priority violations in the new path is less than or equal to that of the original path $V_{new} \leq V_{old}$. This optimization process repeats until no further improvements can be made.

4.2. Stage 2: makespan-oriented optimization via adaptive large neighborhood search

Stage 2 further optimizes the initial solution generated in Stage 1, aiming to minimize the total rescue makespan and the number of priority violations. In this stage, an improved Modified Dynamic Adaptive Large Neighborhood Search (M-D-ALNS) algorithm is proposed based on the ALNS metaheuristic framework. Four targeted destroy operators and two repair operators are designed, which are selected via a roulette wheel strategy. An adaptive weight update mechanism is employed to favor operators that perform better in prior iterations. In addition, a simulated annealing (SA) acceptance criterion is introduced to avoid local optima. We also propose a dynamic destruction rate strategy and a novel weight bonus mechanism to guide the algorithm from global exploration to local exploitation. The procedure of the algorithm is described in Algorithm 1.

4.2.1. Operator library

The efficacy of the ALNS framework is contingent upon its operator library, we customize six operators tailored to the characteristics of the PD-DRRP problem, as illustrated in, as illustrated in Fig. 7.

A. Destroy Operators **Worst Removal** (Fig. 7a): For each task i in the current solution, removing i yields S_{new} with benefit

$$\text{Benefit}_i = F(S_{current}) - F(S_{new}) \quad (23)$$

Tasks are sorted by Benefit_i and those with the largest values (high detour or penalty) are removed first, directly eliminating bottleneck tasks.

Related Removal (Fig. 7b): Inspired by Wang et al. [52], we define relevance between tasks i and j by combining spatial and priority proximity:

$$R(i, j) = w_d \cdot d_{ij} + w_p \cdot |\text{priority}(i) - \text{priority}(j)| \quad (24)$$

Here, d_{ij} is the normalized Euclidean distance, and the second term is the normalized difference in priority rank. A seed task is chosen, and tasks with low relevance to it are removed to reconstruct the local region or priority layer.

Violation Removal (Fig. 7c): This operator scans adjacent task pairs (i, j) sequentially along each route. A priority inversion is identified if task i is visited before j , while $\text{priority}(i) > \text{priority}(j)$. Upon detecting such violations, nodes with higher priority that should be scheduled earlier are preferentially removed. All such nodes are then sorted, and the n tasks involved in the most severe violations are removed. This operator is specially designed for priority constraint optimization in this study.

Random Removal: Randomly removes a given number of tasks from all routes without using structural information, providing diversification and helping the search escape local optima.

B. Repair Operators **Greedy Insertion** (Fig. 7d): This operator performs a two-stage insertion to balance solution quality and computational cost.

Phase 1 Screening: For each task to be inserted, roughly assess its incremental cost (travel time + estimated penalty change) at all possible insertion positions, and retain the position with the lowest cost.

Phase 2 Refinement: The operator selects the top- k (e.g., $k = 3$) most promising candidate positions from Phase 1. Only for these k candidate positions, the objective function is precisely recalculated, and the optimal position is finally selected to complete the insertion.

This strategy can control computational costs while ensuring satisfactory solution quality.

Algorithm 1

Algorithm 1: M-D-ALNS Pre-optimization

Input: Initial solution s_0 ; objective $F(\cdot)$; max iterations N_{iter} ; initial temperature T ; cooling rate γ ; response factor r ; update period L ; rewards $(\sigma_1, \sigma_2, \sigma_3)$.

Output: Best static plan s_{best} .

```

1  $s_{cur} \leftarrow s_0$ ;  $s_{best} \leftarrow s_0$ ;
2 Initialize operator weights  $\{w_i\}$ , scores  $\{S_i \leftarrow 0\}$  and uses  $\{U_i \leftarrow 0\}$ ;
3 for  $k = 1$  to  $N_{iter}$  do
4    $progress \leftarrow k / N_{iter}$ ;  $\theta \leftarrow 0.25(1 - progress) + 0.05 progress$ 
5   Select a destroy operator  $d$  and a repair operator  $q$  by roulette based on  $\{w_i\}$ ;
6    $\hat{s} \leftarrow d(s_{cur}, \theta)$ ;  $s_{new} \leftarrow q(\hat{s})$ ;
7    $\Delta F \leftarrow F(s_{new}) - F(s_{cur})$ ;
8   if  $\Delta F < 0$  then
9      $s_{cur} \leftarrow s_{new}$ ;
10    if  $F(s_{new}) < F(s_{best})$  then
11       $s_{best} \leftarrow s_{new}$ ;  $S_d, S_q \leftarrow \sigma_1$ ;
12    else
13       $S_d, S_q \leftarrow \sigma_2$ ;
14  else if  $\exp((F(s_{cur}) - F(s_{new})) / T) > \text{rand}(0, 1)$  then
15     $s_{cur} \leftarrow s_{new}$ ;  $S_d, S_q \leftarrow \sigma_3$ ;
16   $U_d, U_q \leftarrow 1$ ;
17   $T \leftarrow \gamma T$ ;
18  if  $k \bmod L = 0$  then
19    foreach operator  $i$  do
20       $w_i \leftarrow \max(0.01, (1 - r)w_i + r \cdot \frac{S_i}{\max(1, U_i)})$ ;
21       $S_i \leftarrow 0$ ;  $U_i \leftarrow 0$ ;
22 return  $s_{best}$ ;

```

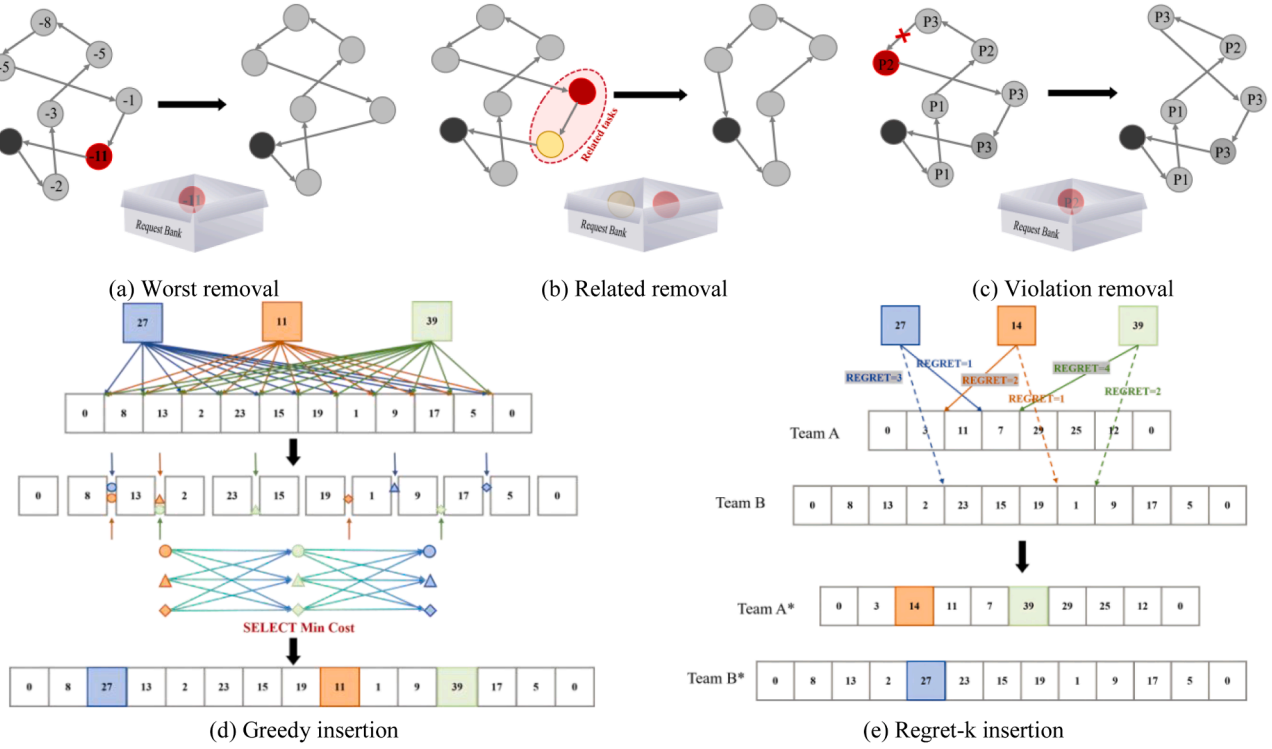


Fig. 7. Schematic diagram of the M-D-ALNS operator library. Top row: destruction operators including the proposed violation removal. Bottom row: insertion operators.

Regret-k Insertion (Fig. 7e): For each task to be inserted, calculate its "optimal insertion cost" $Cost1(i)$ and its "second-best insertion cost" $Cost2(i)$. Define regret value:

$$Regret(i) = Cost2(i) - Cost1(i) \quad (25)$$

Instead of prioritizing the smallest task, it prioritizes inserting the task with the highest regret value. This approach alleviates the short-sightedness of pure greedy insertion.

4.2.2. Adaptive control mechanisms and SA acceptance

The M-D-ALNS framework adopts a dynamic destruction rate strategy, whereby the destruction proportion linearly decreases from 25% to 5% as the iteration proceeds. A higher destruction rate is used in the early stage to strengthen global exploration, while a lower rate is applied in the later stage for focused local exploitation. This strategy is experimentally verified to significantly improve algorithm performance, as shown in Table S3.

Taking the improvement contribution to the objective function as the evaluation metric, the algorithm continuously records and quantifies the performance of each operator throughout iterations. The response factor r is used to adjust the retention ratio of historical weights and the weight proportion of normalized operator scores in the current cycle. Based on the cumulative reward scores and actual usage counts of operators within each cycle, their weights are dynamically adjusted via weighted iteration formulas, with a lower bound imposed on weights to preserve the exploratory capability of operators.

$$w_{i,new} = \max\left(0.01, (1-r) \cdot w_{i,old} + r \cdot \left(\frac{S_i}{U_i}\right)\right) \quad (26)$$

Where S_i represents the cumulative reward score of operator i in the current cycle, which is calculated by the superposition of σ_1 , σ_2 and σ_3 according to the optimization effect. U_i denotes the actual number of uses of operator i in the current cycle, which is used to normalize the score and avoid evaluation bias caused by differences in usage frequency.

Simultaneously, a differentiated multi-level reward system is established during weight updates: On one hand, it maximizes the contribution weight of the global optimum solution by superimposing reward strategies ($\sigma_1 + \sigma_2$). On the other hand, it innovatively sets the reward coefficient for "suboptimal solutions accepted by the Metropolis criterion" (σ_3) significantly higher than that for "conventional locally optimal solutions" (σ_2). This non-monotonic reward mechanism aims to disrupt greedy search paths, granting higher weights to operators capable of "escaping local extrema." Combined with the probabilistic acceptance characteristic of simulated annealing, it effectively prevents premature convergence, achieving deep synergy among operator performance tracking, dynamic weight adjustment, and global optimization mechanisms.

To avoid being trapped in local optima, the ALNS employs a Simulated Annealing (SA) acceptance criterion. New solutions that are better than the current one are always accepted. Worse solutions may still be accepted with a probability that decreases over time:

$$P(\text{accept}) = \exp\left(\frac{F(S_{current}) - F(S_{new})}{T}\right) \quad (27)$$

where T is the temperature parameter, which is gradually decreased according to the cooling schedule.

4.3. Stage 3: agile dynamic response

In actual disasters, the estimated demand and service time often contain deviations, and the actual number of trapped people at affected nodes can only be gradually confirmed when each team performs search and rescue tasks [53]. In the execution phase, we draw on the Discrete Event Simulation (DES) [54] method and employ an event-driven strategy to handle uncertainties such as service time surges. Re-planning is triggered strictly in accordance with the "Impact Gap" criterion, as presented in Algorithm 2.

If complete re-planning is executed whenever new information is acquired, both computational and command burdens become prohibi-

Algorithm 2

Algorithm 2: Event-Driven Dynamic Response Loop**Input:** Pre-plan π_{static} ; actual data L_{act} ; predicted data L_{est} ; thresholds $c_1, c_2, \zeta, \Delta t_{\min}, H_K$.**Output:** $T_{\text{final}}, N_{\text{replan}}$.

```

1 Initialize event queue  $Q$  (first Arrival per team),  $t \leftarrow 0$ ,  $t_{\text{last}} \leftarrow -\infty$ ,  $N_{\text{replan}} \leftarrow 0$ ;
2  $T_{\text{base}} \leftarrow \text{CalcMakespan}(\pi_{\text{static}}, D_{\text{est}})$ ;
3 while  $Q \neq \emptyset$  do
4   Pop earliest event  $(t, \text{type}, k, i)$ ;
5   if  $\text{type} = \text{Arrival}$  then
6     Observe  $s_i^{\text{act}}$ ; push Completion at  $t + s_i^{\text{act}}$ ;
7   else if  $\text{type} = \text{Completion}$  then
8     Update  $H_k$  with  $\Delta = s_i^{\text{act}} - s_i^{\text{est}}$ ; compute  $T_{\text{curr}}$ ;
9     // Triggers (any true) + cooling
10     $\text{trig} \leftarrow (s_i^{\text{act}} > (1 + c_1)s_i^{\text{est}}) \vee (s_i^{\text{act}} < (1 - c_2)s_i^{\text{est}}) \vee (T_{\text{curr}} > (1 + \zeta)T_{\text{base}}) \vee \text{Unidirectional}(H_k, m)$ ;
11     $\text{cooled} \leftarrow (t - t_{\text{last}} \geq \Delta t_{\min})$ ;
12    if  $\text{trig} \wedge \text{cooled}$  then
13      Freeze ongoing/en-route tasks; extract unvisited tasks  $N_{\text{pend}}$ ;
14       $\pi \leftarrow \text{ShortHorizonALNS}(N_{\text{pend}}, D_{\text{est}})$ ; rebuild  $Q$ ;
15       $t_{\text{last}} \leftarrow t$ ,  $N_{\text{replan}} \leftarrow N_{\text{replan}} + 1$ ,  $T_{\text{base}} \leftarrow T_{\text{curr}}$ ;
16      Schedule next task  $j$  for team  $k$ ; if exists, push next Arrival event;
17  $T_{\text{final}} \leftarrow t$ ; return  $T_{\text{final}}, N_{\text{replan}}$ ;

```

tive. Conversely, the absence of re-planning leads to severe inefficiency. To address this trade-off, this paper proposes a Bidirectional Event-Driven Response Mechanism Based on Local and Global (BEDRM-LG). This mechanism integrates active opportunity capture and passive risk defense. After each task is completed, the strategy evaluates the following multi-dimensional triggering conditions:

1. **Local deterioration:** when the s_i^{act} is significantly greater than the s_i^{est} , indicating a severe single-point bias, formulated as:

$$s_i^{\text{act}} > (1 + c_1)s_i^{\text{est}} \quad (28)$$

Where s_i^{act} and s_i^{est} are derived from L_{act} and L_{est} respectively as defined in Section 3.3.2.

2. **Global deterioration:** The current makespan, calculated based on the actual rescue duration of visited nodes and the estimated duration of unvisited nodes, is significantly worse than the baseline rescue time of the initial pre-plan:

$$T_{\text{new}} > (1 + \zeta)T_{\text{old}} \quad (29)$$

3. **Active optimization:** If the s_i^{act} is significantly earlier than s_i^{est} , it indicates that the affected node has released substantial time resources, which can be used to accept new urgent tasks or reallocate tasks to other teams delayed by disruptions.

$$s_i^{\text{act}} < (1 - c_2)s_i^{\text{est}} \quad (30)$$

4. **Continuous fluctuation:** To capture systematic deviations in team operational efficiency, the strategy maintains a time deviation sequence $H_k = \{\Delta_1, \Delta_2, \dots, \Delta_m\}$ or the most recent m tasks of each team k , where $\Delta_i = s_i^{\text{act}} - s_i^{\text{est}}$. Re-optimization is triggered if and only if all deviations in the sequence are unidirectional.

$$\text{Trigger}_{\text{drift}} = (\forall \Delta \in H_k, \Delta > 0) \vee (\forall \Delta \in H_k, \Delta < 0) \quad (31)$$

To ensure the robustness of the re-response mechanism in complex

rescue scenarios, sensitivity analyses are conducted on the above thresholds, as presented in Table S4. In this study, c_1 and c_2 are set to 0.3, ζ to 0.15, and H_k to 2.

When any of the triggering conditions is met, the short-cycle M-D-ALNS optimization framework will be applied to re-plan the flight path for the unfinished tasks:

1. Completed tasks and their actual service times remain unchanged, and re-planning is only performed for unvisited nodes based on their estimated information.
2. The current position and current simulation time of each rescue team are taken as the new initial state, and tasks being executed or en route cannot be interrupted.
3. The unvisited portion of the original plan is replaced by the newly generated schedule for the remaining tasks.

5. Computational experiments

This section presents a series of computational experiments to systematically evaluate the contribution of each stage. All experiments are carried out on constructed Solomon benchmark instances, adapted to scenarios involving post-disaster road damage and uncertain rescue demands. The proposed framework is implemented in Python 3.8.3 and run on a computer equipped with an Intel(R) Core(TM) i7-10700K CPU @ 3.80 GHz and 64 GB of RAM.

To verify the validity and scalability of the proposed framework, we conducted experiments on three types of typical distribution characteristics of the Solomon benchmark instances: the C series (clustered distribution), the RC series (partly clustered + partly dispersed), and the A/R series (random distribution), covering multi-scale instances with 25 to 200 nodes. To facilitate clear presentation and in-depth analysis, unless otherwise specified, this article adopts the medium-scale RC-n100-k5-m10 instance as the representative case for all visualizations and detailed discussions in the main text. The detailed data of the example can be found in Table S5. The common parameter settings for all experiments are shown in Table 3.

5.1. Performance of the pre-optimized solution framework

To evaluate the effectiveness of the pre-optimized planning components, we first generate an initial solution using the Stage 1 methodology. This solution is then optimized using the Stage 2 M-D-ALNS

Table 3
Core experimental parameters.

Parameter Name	Value
Number of iterations	2000
Initial temperature (T_{start})	10
Final temperature (T_{end})	0.01
Cooling rate	0.999
Weight update cycle	100
Reaction factor r	0.2
$\sigma_1, \sigma_2, \sigma_3$	33, 9, 13
Number of rescue teams (m)	5
Rescue team speed(km/h)	40

framework. The primary metric for comparison is the objective function value. Table 4 reports the detailed results of each rescue team in this instance. It can be seen that the initial path generated in stage 1 has almost no priority violations. After optimization by ALNS, the maximum rescue time has been reduced by approximately 22.2 h, and the rescue efficiency has increased by 31.16%.

Table S6 details the responsible areas, rescue points, rescue routes, and completion times for each team during the experiments. Fig. 8 compares the spatial distribution of the initial and optimized routes. The initial plan in Fig. 8a already shows clear zoning and largely follows the priority rules. The optimized plan in Fig. 8b further removes unnecessary retracing and balances the workload among teams.

As can be seen from the ALNS convergence curve in Fig. 9, the objective value drops rapidly during the first 110 iterations, where most of the improvement is achieved. Afterwards, the curve flattens and the gap between the best and current solutions narrows. The high-quality initial solution thus provides a strong starting point, enabling major gains in a short time, while later iterations focus on local fine-tuning to explore slightly better solutions.

To further examine the mechanism of high-quality initial solutions on convergence performance, this study selects typical small-to-medium (RC-n100) and large-scale (RC-n200) instances. Using multiple random seeds and path refinement perturbations with varying intensities, 30 independent initial solution samples are generated for each instance, covering a gradient from high to low compliance. Given that initial violations are discrete counts, convergence iterations exhibit a long-tailed distribution, and their relationships are nonlinearly monotonic, the distribution-free Spearman's rank correlation coefficient is adopted to quantify the effects of initial violation counts and path structure (makespan) on convergence speed, defined as the number of iterations required to reach within 105% of the global optimal solution (as shown in Fig. 10, Fig. 11).

Experimental results reveal the decisive influence of problem scale on convergence behavior: at small and medium scales, all correlation tests are insignificant $p > 0.05$, demonstrating the algorithm's exceptional robustness in low-dimensional spaces to effectively correct sub-optimal initial solutions. However, in large-scale cases ($n=200$), the initial violation count exhibits a strong positive correlation with convergence time ($\rho_{100}^{viol} \approx 0.913, p < 0.0001$), indicating that local extremum traps formed by violation penalties in high-dimensional spaces significantly deplete search resources. Conversely, initial makespan exhibits a significant negative correlation with convergence time

Table 4
Comparison of solution quality before and after stage 2.

Metric	Stage 1 initial solution	Stage 2 optimized solution	Improvement (%)
Makespan (hours)	74.93	52.76	31.16%
Total Priority Violations	1	0	100%
Objective Value	1074.93	52.76	95.09%
Total Distance Traveled	1091.76	1532.60	-40.38%

($\rho_{100}^{mksp} \approx -0.653$), indicating that short-sighted distance-greedy optimization shortens initial paths but incurs high repair costs. This result confirms that as solution space complexity increases exponentially, relying solely on an algorithm's adaptive repair capability struggles to overcome the "penalty barrier." It thus strongly argues for the necessity of incorporating priority-aware construction strategies to generate high-quality initial solutions in large-scale scenarios.

5.2. Heuristic benchmark comparison

To systematically evaluate the performance of the proposed M-D-ALNS and validate the effectiveness of its core mechanisms, this section presents a comprehensive validation scheme consisting of ablation experiments and comparisons with baseline algorithms. To quantify the contribution of the multi-level improvement strategies, a basic ALNS is constructed as a control algorithm. The baseline removes the adaptive weight mechanism, dynamic destruction strategy, dedicated operators, and hybrid heuristic initialization strategy proposed in this paper, retaining only the fundamental random removal, worst removal, and greedy insertion operators with an equal-probability selection mechanism. This design establishes a "zero baseline" to highlight the improvements of the novel operators and adaptive mechanisms in handling complex constraints and exploring the solution space.

To ensure comprehensive and cutting-edge comparisons, this paper references improved strategies for constrained task assignment and path planning problems in relevant fields. By adapting their encoding/decoding methods, constraint handling mechanisms, and discretization strategies, we developed the Improved Genetic Algorithm (IGA) [55], Improved Particle Swarm Optimization (IPSO) [56], Improved Ant Colony Optimization (IACO) [57], and Improved Differential Evolution (IDE) [58]. These enhanced versions integrate the priority constraint handling and path optimization mechanisms validated in the literature, ensuring the benchmark algorithms possess the capability to handle the complex constraints addressed in this paper.

Due to significant heterogeneity in population structure, search mechanisms, and computational load per iteration across different algorithms, traditional termination criteria based on a fixed number of iterations struggle to accurately reflect actual computational efficiency differences. Therefore, in accordance with the requirements of China's "National Earthquake Emergency Response Plan", this paper uniformly adopts a maximum runtime of 3600 seconds as the termination condition. This optimization runs concurrently with the unavoidable physical mobilization phase, ensuring that the computation does not delay actual rescue operations. Each algorithm was run independently 10 times to mitigate random interference. Welch's t-test was employed for statistical significance analysis of experimental results.

Table 5 presents a performance comparison of the proposed M-D-ALNS algorithm against base ALNS and four improved benchmark algorithms IACO, IDE, IGA, IPSO on the same test case. Given the case's high priority violation penalty coefficient of 1000, the objective function value exhibits extreme sensitivity to constraint satisfaction, significantly amplifying differences in algorithmic robustness and optimization capability. Results demonstrate that M-D-ALNS exhibits a dominant lead in optimization capability, with an average value 54.44 substantially lower than all other algorithms (446+), and an optimal value 51.07 approaching the theoretical lower bound. Despite the severe priority violation penalty of 1000, M-D-ALNS consistently yields results well below 1000, proving its ability to successfully find feasible solutions satisfying strict priority constraints in nearly all runs. While the base ALNS occasionally found good solutions (Min 54.19), its maximum value reached 1056.12 with a standard deviation of 499.10. This indicates extreme instability without specialized operators, frequently leading to constraint violations due to getting stuck in local optima. The four improved benchmark algorithms all yielded average values between 446 and 676, with Gap% exceeding 770%. This indicates that

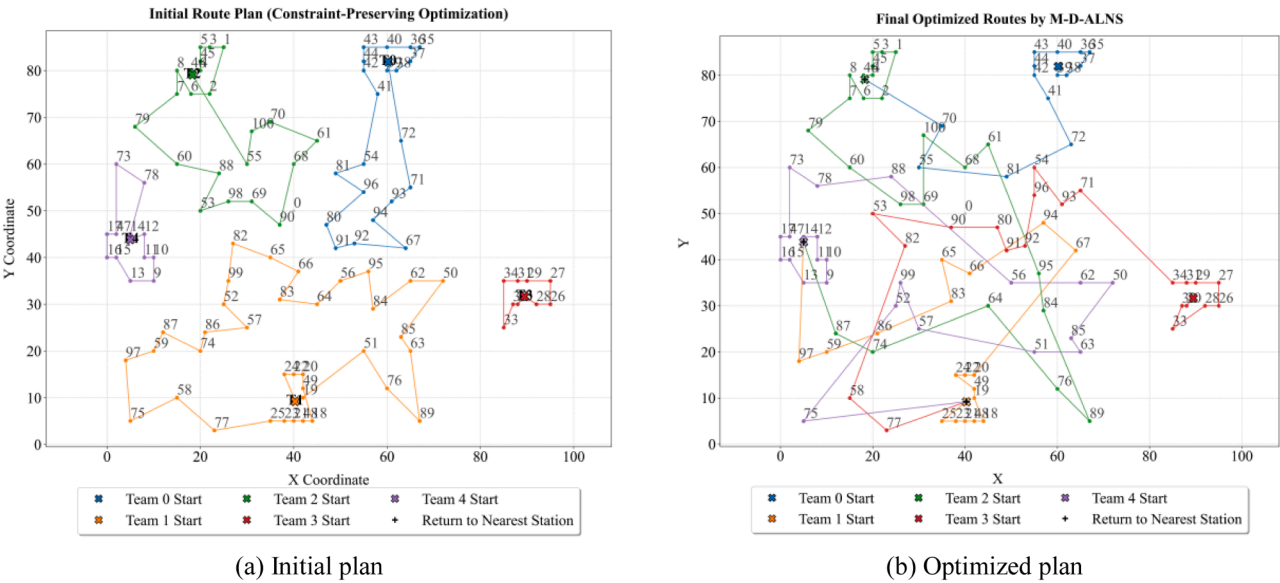


Fig. 8. Comparison of rescue team routes before and after ALNS optimization.

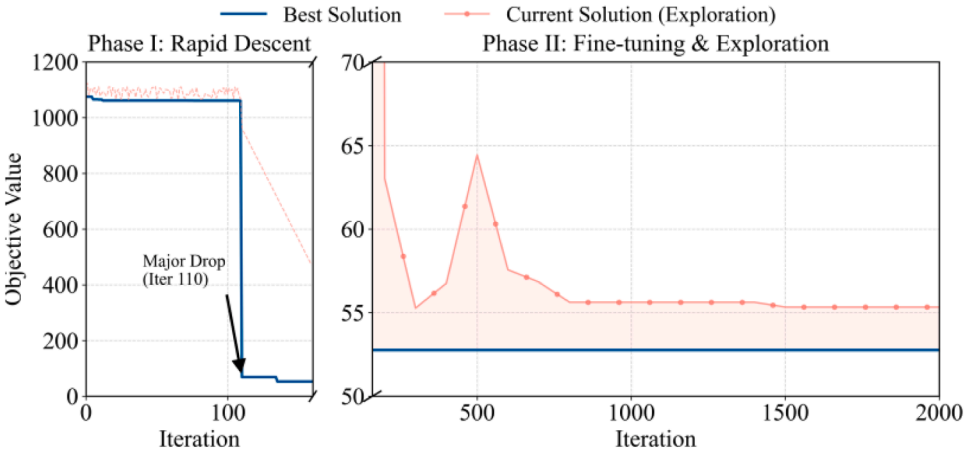


Fig. 9. Convergence curve of the ALNS.

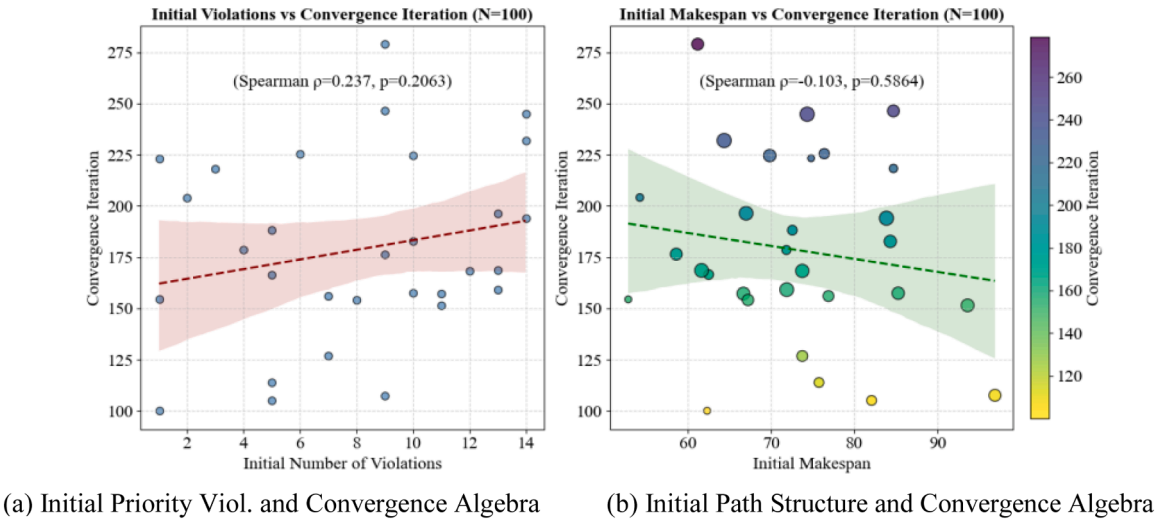


Fig. 10. (n = 100) Correlation analysis between initial Solution compliance and convergence speed.

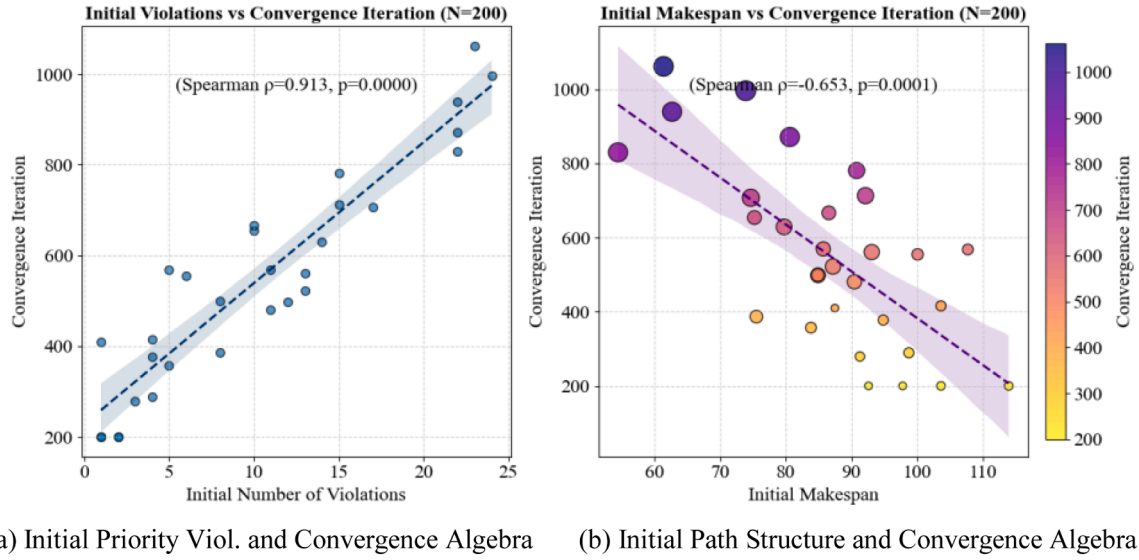


Fig. 11. (n = 200) Correlation analysis between initial Solution compliance and convergence speed.

Table 5

Comparison of pre-planning performance under different heuristic benchmarks.

Algorithm	Min	Max	Mean	Std	Avg.Imp%	Gap%
ALNS	54.19	1056.12	556.09	499.10	48.27	988.79
IACO	283.96	564.14	446.31	76.35	58.48	773.85
IDE	589.10	774.92	676.71	53.10	37.05	1224.95
IGA	382.51	564.69	475.67	65.32	55.75	831.33
IPSO	368.12	545.05	460.00	56.64	57.21	800.66
M-D-ALNS	51.07	61.39	54.44	2.71	94.94	6.60

under strongly constrained environments, despite enhancements and the introduction of homogeneous local search, the improved benchmark algorithms still struggle to converge to the global optimum when handling the coupled problem of strict priority constraints and path optimization. They often get stuck in local optima with partial soft constraint violations or extremely circuitous paths.

The M-D-ALNS algorithm demonstrates exceptional solution quality and robustness for high-constraint, high-penalty disaster relief task assignment problems, achieving an average improvement rate of 94.94% and an internal variation Gap% of only 6.60%. The stark contrast in ablation experiment data directly quantifies the necessity of the proposed improvement strategy.

5.3. Comparison of initialization strategies

To isolate the value of our strategic initialization, we compare five different strategies for assigning tasks and building the initial solution:

- HGA (Proposed): The hierarchical greedy assignment strategy presented in Section 4.1.
- Greedy Priority Assignment (GPA): A priority-oriented scheduling rule that sorts affected nodes in descending order of priority and sequentially assigns each node to the nearest team to prioritize high-demand regions [59,60].
- Priority-Aware Load-Balancing Assignment (PA-LBA): A weighted load is computed for each node using estimated trapped population and priority rank. Nodes are sorted in descending order of weighted load and assigned iteratively to the team with the minimum current load, balancing workload and reducing priority violations [61].
- Geographic Nearest-Neighbor Assignment (GNNA): A pure distance-based rule that assigns each node to the closest team regardless of priority or load balance [62].

- Sweep Algorithm Assignment (SAA): Nodes are sorted by polar angle around the epicenter and partitioned into M angular sectors, each assigned to one team. This method forms fan-shaped service zones and improves route continuity [63,64].

For a fair comparison, all strategies use the same Stage 2 ALNS optimizer, with the same number of iterations and parameters. The only difference is the initialization strategies.

The results show that in all test cases, the HGA strategy can generate solutions with the lowest initial objective value in the shortest time. Before any metaheuristic search, it has already achieved the minimum number of priority violations. Especially in clustered cases, the number of initial violations is extremely low (0 or 1), indicating that the initialization stage has basically met the requirements of humanitarian rescue.

After applying the M-D-ALNS optimizer to all initial solutions, the HGA strategy still achieves the lowest or near-lowest makespan in almost all cases and converges faster to a stable plateau. Other initialization strategies require much longer optimization time to approach a similar level and may even converge to slightly inferior local optima. This is because the high-quality initial solutions generated by HGA significantly reduce the computational burden on ALNS operators, avoiding the time-consuming process of continuously repairing a large number of priority violations. The actual runtime T_{gen} (See Table S8, it summarizes the results on multiple modified benchmark instances) demonstrates that the HGA strategy is indeed superior in both search efficiency and actual computational cost.

Fig. 12 presents the iteration curves of the objective function under each strategy. All curves decrease over time, but the curve of HGA starts at a much lower value and maintains a leading position throughout.

Under the unified M-D-ALNS optimization framework, regardless of the instance scale and characteristics, all different initialization strategies ultimately achieve nearly complete elimination of constraint violations, with the optimization rate of the objective function value exceeding 90% compared to the initial solutions. This result fully verifies the generalization and effectiveness of the ALNS framework proposed in this study.

5.4. Evaluation of the dynamic response framework

To quantitatively validate the effectiveness of the Event-Driven Dual-Trigger Mechanism (BEDRM-LG), this paper designs two sets of Monte Carlo simulations with different levels of information visibility:

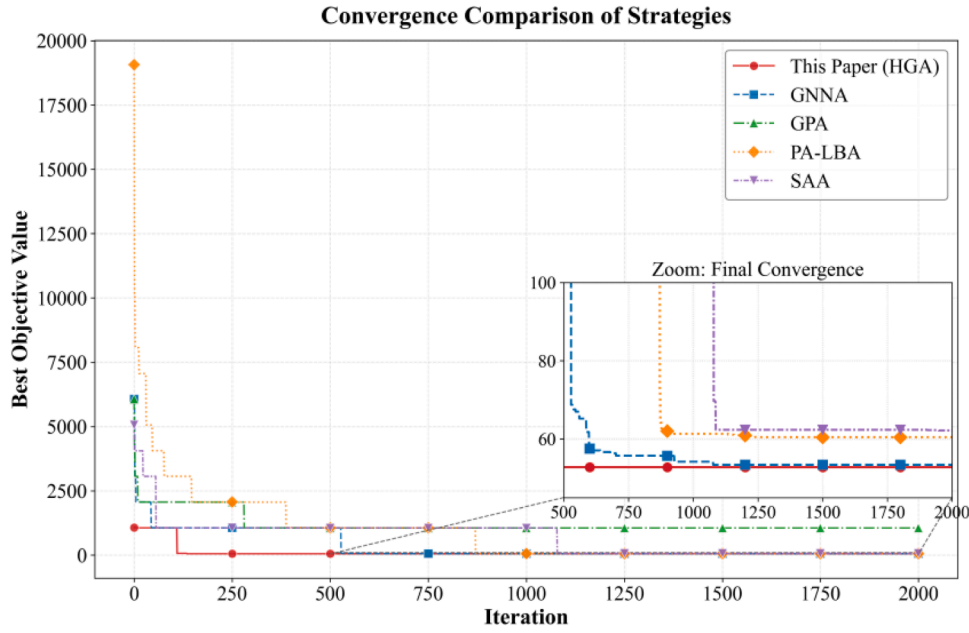


Fig. 12. Convergence of ALNS under different initialization strategies (100-node example).

1. Realistic dynamic strategy: The BEDRM-LG under a real-world information perspective, which performs re-optimization when the triggering conditions are satisfied. Re-planning optimizes the remaining routes based on estimated information.

2. God View dynamic strategy: Serving as the theoretical performance upper bound, it adopts the same triggering rules as the realistic strategy but assumes access to the true information of unexecuted nodes during re-planning. This strategy is infeasible in practice and is only used as a benchmark to quantify the performance loss of BEDRM-LG under incomplete information.

Both strategies use the same set of data sources, L_{act} and L_{est} . To avoid experimental bias caused by single-scenario tests, both normal error scenarios and black-swan extreme error scenarios are constructed. Multiple repeated simulation runs are conducted and compared with the pre-planning benchmark that only uses estimated data.

(1) Normal-error scenario. This scenario simulates the ubiquitous deviation in rescue information. Based on the estimated service time L_i^{est} , a random perturbation factor ϵ_i following a truncated normal distribution is introduced: $\epsilon_i \sim N(1.0, 0.1^2)$, $\epsilon_i \in [0.8, 1.2]$. The actual service time is updated as:

$$L_i^{act} = \text{round}(L_i^{est} \times \epsilon_i) \quad (32)$$

(2) Black Swan scenario. This scenario simulates local extreme secondary disasters. On the basis of the normal error, 10% of nodes in set V are randomly selected as the severely affected area $\Omega_{disaster}$, and a severe demand amplification factor $\gamma_i \sim U(3.0, 5.0)$ is imposed:

$$L_i^{act} = \begin{cases} L_i^{est} \times \gamma_i, & i \in \Omega_{disaster} \\ L_i^{est} \times \epsilon_i, & i \notin \Omega_{disaster} \end{cases} \quad (33)$$

Table 6 summarizes the optimization performance of the BEDRM-LG strategy on the case from both a realistic perspective and an omniscient perspective (average over 20 simulations), as well as its optimization

Table 6

Performance comparison of dynamic strategies under uncertainty from the perspective of reality and the perspective of God.

Scenario	Strategy	Avg. makespan (h)	Avg. Imp (%)	Std
Actual-Data	Pre-Op	53.00	/	/
	Realistic	41.84	21.06%	2.36
	God View	44.63	15.79%	2.97
Normal-Error	Pre-Op	51.98	/	/
	Realistic	44.31	14.72%	6.49
	God View	44.16	15.03%	6.22
Black Swan	Pre-Op	77.52	/	/
	Realistic	57.02	26.09%	6.68
	God View	59.88	22.14%	8.33

performance under multiple simulation scenarios.

Upon examining the experimental data, it was observed that in certain simulation runs (details of which can be found in Table S7), the actual dynamic strategy performed even better than the omniscient upper limit. It is traditionally believed that the God View strategy, which possesses perfect information of the future, should guide the algorithm to converge to a better theoretical lower bound. However, experimental data demonstrate that under a limited number of iterations (consistent with the timeliness requirements of rescue operations), the Realistic strategy with limited information ultimately yields a lower global makespan than the God View strategy with perfect information. This phenomenon arises because real-world disaster data are contaminated by high-frequency noise and outliers. The God View strategy attempts to exploit its complete information by fitting every noisy data point using heuristic algorithms, leading to overfitting. To accommodate certain local extreme observations, it disrupts the well-structured global rescue topology, ultimately resulting in a systematic loss of efficiency. These findings validate the effectiveness of the BEDRM-LG framework proposed in this study: in the critical and highly uncertain early phase post-disaster, agility and robustness are far more critical than theoretical optimality.

5.5. Comparison of efficacy of the dynamic response mechanism

In the final experiment, we compared the performance of several commonly used dynamic response strategies. All methods adopted the

same baseline scheme generated in the second stage, and the road damage degree and actual number of people to be rescued at all task nodes in the same comparative experiment were generated with the same random seed, the objective physical attributes of the environment remained unchanged regardless of the scheduling strategy employed.

The benchmark strategies are:

- **A_Static:** No re-planning is performed; serves as the reference baseline.
- **B_Periodic:** Re-optimizes at fixed time intervals [65]. Based on the conventional engineering practices for disaster area dispatching, the interval period is set at 5 hours.
- **C_RollingHorizon:** Re-optimizes after completing a fixed number of tasks, following a task-based rolling horizon scheme similar to dynamic VRP approaches [29]. Specifically, in this experiment, the duration for completing the task was set at 10.
- **D_RealisticTrigger (proposed):** The BEDRM-LG strategy under a realistic information perspective..
- **E_GodViewTrigger:** The BEDRM-LG strategy under an omniscient information perspective.

Table 7 summarizes the average simulation results of each strategy across three post-disaster scenarios: real data, normal error, and black swan events. Overall, all dynamic strategies significantly reduce the actual makespan compared to the static pre-planning baseline (A_Static), and the improvement rate exhibits an upward trend as the number of re-planning operations increases. As illustrated in Fig. 13, under the normal error scenario, the BEDRM-LG strategy takes the lead in performance, already achieving near-theoretical optimal optimization results. In the black swan scenario, the makespan of all strategies rises due to extreme uncertainty; however, the BEDRM-LG strategy outperforms the GodView strategy approach with superior stability.

However, the return on computational investment (i.e., re-planning efficiency) varies considerably across different triggering mechanisms. As shown in Fig. 14, under the normal error scenario, the single re-planning efficiency of the task-rolling strategy is merely 2.57 h per operation; whereas in the real-data scenario, the efficiency of the periodic strategy reaches only 3.13 h per operation. The triggering mechanisms of these strategies rely on fixed time windows or task quantity thresholds, which are mostly set based on empirical or subjective judgment and thus lack the capability to objectively respond to the dynamic evolution of disaster conditions. When the triggering timing deviates substantially from the actual disaster situation, the re-planning algorithm may be invoked blindly when the disaster situation does not significantly deteriorate, leading to unnecessary computational waste. Conversely, if the optimization process is initiated only after the disaster has severely worsened, the optimal intervention window will be missed, resulting in a distinct diminishing marginal return of the optimization performance.

In contrast, the proposed BEDRM-LG strategy maintains leading single re-planning efficiency across all scenarios. Through the on-demand intervention mechanism, each re-planning solution can

accurately address the critical delay bottlenecks of the system, thereby avoiding unnecessary computational overhead and achieving a higher return on computational investment.

6. Conclusions

This paper proposes a three-stage decision-making framework for post-disaster rescue, aiming to address two critical issues in practical rescue operations: information uncertainty and humanitarian priority in rescue.

To ensure that the research remains grounded in real post-disaster rescue practice, this study models the actual disaster-affected environment. First, an improved K-means method based on the silhouette coefficient is employed to partition the disaster area into multiple rescue zones. To simulate post-earthquake information uncertainty, the actual casualty numbers are generated using a density-based Poisson model, and noisy estimated values are produced by adding Gaussian errors. Considering road damage after earthquakes, a concentric circle road damage model is constructed, and travel times are increased according to the severity of road deterioration. On this basis, three indicators are defined in line with humanitarian rescue requirements: total life value, disaster impact, and rescue efficiency density. The entropy weight method is then used to construct an objective priority matrix for all zones.

This paper designs a two-step pre-optimization method. In the task allocation stage, a priority-driven greedy hierarchical assignment strategy is adopted to ensure that each team is assigned a compact responsibility zone containing high-priority tasks evenly. During pre-planning of rescue schemes, a hybrid heuristic is proposed that integrates a priority-aware nearest neighbor algorithm and a priority-preserving 2-opt local perturbation, generating open routes that strictly follow the priority order. Experiments demonstrate that this strategy significantly improves the convergence speed of the algorithm in the global optimization stage.

Subsequently, an modified dynamic adaptive large neighborhood search (M-D-ALNS) is proposed as the global optimizer. Four destruction operators and two repair operators are designed for the makespan and constraint violations of rescue schemes. A weight adaptive update mechanism and a dynamic destruction strategy are introduced to further compress the makespan of pre-planned solutions and reduce access violations.

In the dynamic re-planning stage, a Bidirectional Event-Driven Response Mechanism Based on Local and Global (BEDRM-LG) is proposed. The M-D-ALNS framework is reused to rapidly re-plan unvisited tasks at re-planning instants.

The effectiveness of the framework is verified through five groups of experiments. First, the performance of the pre-planning framework is examined. The proposed M-D-ALNS algorithm can obtain optimal solutions within 200 iterations and outperforms other benchmark algorithms. Through correlation analysis between initial solution feasibility and algorithm convergence speed, it is found that M-D-ALNS exhibits strong robustness on medium- and small-scale instances. For large-scale

Table 7

Simulation results of different dynamic strategies under multiple scenarios (average over 10 Monte Carlo simulations).

Policy	Actual Data				Normal-error				Black Swan			
	T_{max}	Rep.	Imp.	Effi.	T_{max}	Rep.	Imp.	Effi.	T_{max}	Rep.	Imp.	Effi.
A_Static	51.70	0	0	0	51.99	0	0	0	77.98	0	0	0
B_Periodic	43.42	8.00	15.90	3.13	42.73	8.20	17.82	2.99	59.77	11.50	23.37	3.97
C_RollingHorizon	44.07	9.00	14.75	2.85	44.04	9.00	15.30	2.57	60.21	9.00	22.80	4.47
D_Realistic	41.25	11.00	20.21	4.30	41.23	11.60	20.71	3.94	53.63	13.80	31.24	5.91
E_God Vew	42.11	12.00	18.54	3.62	39.90	12.50	23.26	4.00	56.77	15.20	27.21	4.67

Note: T_{max} represents the average maximum completion time, Rep. represents the average number of re-planning times, Imp. The average improvement rate for indicating the completion time is Effi. The average efficiency of a single re-planning is represented by Effi.

Table 8

Significance test for the difference in improvement rates between the BEDRM-LG strategy and other strategy options.

Compared Strategy	Normal-Error Scenario			Black Swan Scenario		
	t-value	p-value	Significance	t-value	p-value	Significance
Periodic	1.7989	0.0888	Marginally significant	3.0347	0.0071	Significant
Rolling Horizon	4.5665	0.0002	Significant	2.2071	0.0405	Significant
God View	-1.7070	0.1050	Not significant	1.7819	0.0916	Marginally significant

The statistical test results presented in Table 8 further corroborate the aforementioned conclusions.

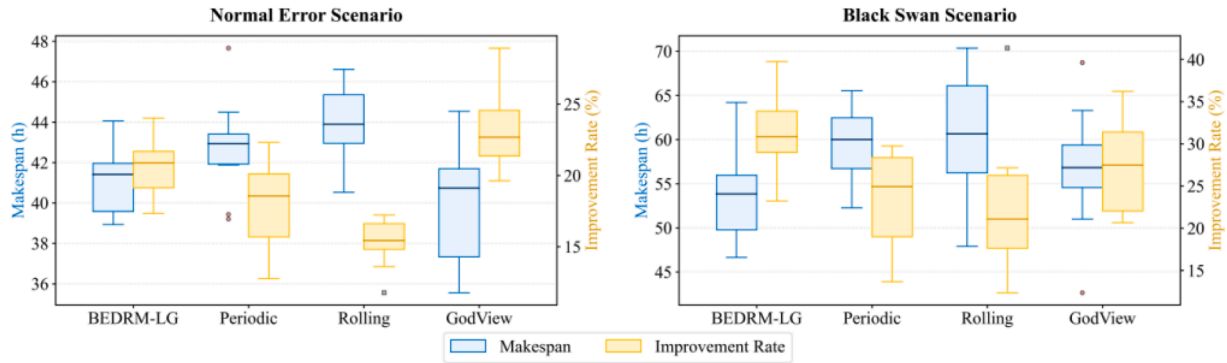


Fig. 13. Core performance comparison of various strategies in different scenarios.

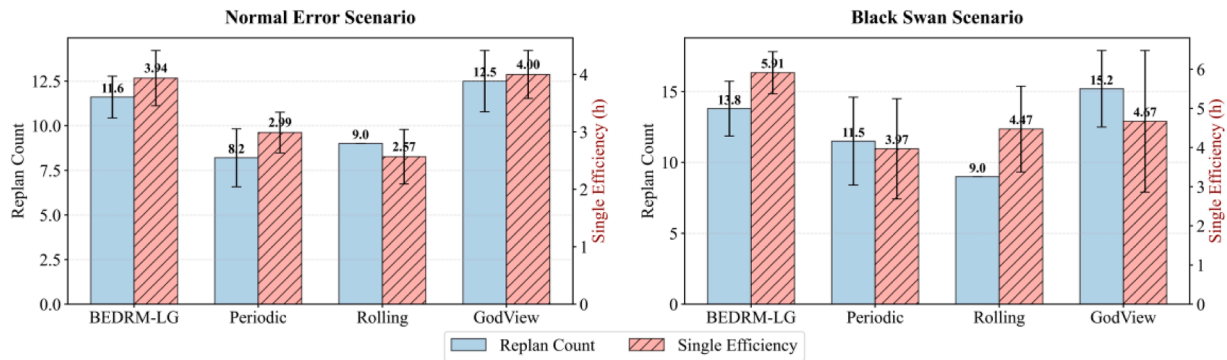


Fig. 14. Comparison of costs and efficiencies of various strategies in different scenarios.

instances, high-quality initial solutions almost determine convergence speed. Therefore, five initialization strategies are further compared. Experiments confirm that the proposed strategy outperforms the priority greedy strategy, high-priority load balancing strategy, geographic nearest strategy, and sweep baseline strategy in both solution quality and computational time. It provides the fastest initial allocation, the lowest initial objective value, the fewest initial constraint violations, and achieves minimal or near-minimal makespan as well as the shortest optimization time after global optimization.

To evaluate the performance of the proposed dynamic response strategy, the BEDRM-LG strategy is compared with the rolling horizon mechanism and periodic re-planning mechanism commonly used in emergency management under various simulation scenarios. A God view benchmark of BEDRM-LG is constructed to quantify its performance loss.

The results show that the BEDRM-LG strategy surpasses all other benchmarks in optimization rate and exhibits stronger robustness. Meanwhile, experiments reveal that perfect foresight does not always yield the shortest makespan, and selective rapid responses may be more valuable than continuous global optimization in highly uncertain environments. Under normal error conditions, the reality-perspective BEDRM-LG achieves a makespan close to the God View benchmark. Under black swan error conditions, its performance even exceeds the

God View benchmark. In deeply uncertain situations, the agility and robustness of strategies are more important than accurate prediction and frequent re-planning.

Therefore, it can be concluded that integrating data-driven zoning planning, multi-indicator priority modeling, metaheuristic optimization, and event-driven re-planning into a unified framework for post-disaster rescue routing is feasible. The framework effectively solves humanitarian rescue problems with strict priority constraints and satisfies the timeliness requirements of emergency response.

However, there are still some limitations. We assume that the rescue teams are homogeneous and have not modeled detailed medical or material restrictions. The damage and demand models are simplified and may not cover all factors in actual situations. Future work can extend this framework to heterogeneous fleets, time windows, multi-period operations, and joint planning of materials and medical resources. Furthermore, this method should also be applied to real disaster data and integrated with real-time information platforms, so that the model and algorithm can provide support for decision-making in actual rescue operations.

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Supplementary Material

The following is the Supplementary data to this article.

CRediT authorship contribution statement

Feng Meining: Conceptualization, Methodology, Software, Data curation, Formal analysis, Writing – original draft, Writing – review & editing, Visualization. **Li Shuanglin:** Conceptualization, Methodology, Writing – review & editing, Resources, Supervision. **Wu Guohua:** Project administration, Supervision, Funding acquisition.

Declaration of competing interest

All authors declare no potential conflicts of interest with any individuals or institutions related to this study.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.orp.2026.100392](https://doi.org/10.1016/j.orp.2026.100392).

Data availability

Data will be made available on request.

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